

# Derrin Bright

Bioinformatics | Biotechnology | ML in Biology

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Portfolio

GitHub

LinkedIn

## Education

**National University of Singapore (NUS)**

Master of Science in Precision Health and Medicine, Yong Loo Lin School of Medicine

Singapore

Aug 2026-Present

**Vellore Institute of Technology (VIT)**

Bachelor of Technology in Biotechnology

**CGPA: 8.8** Vellore, India

2022-2026

## Technical Skills

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|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Computational Biology Tools</b> | BLAST   AlphaFold   AutoDock Vina   GROMACS   MGLTools   Bioconductor   OpenBabel   Cytoscape   Seurat   GSEA   FIMO                                                         |
| <b>Data Analysis Tools</b>         | Neo4J   KNIME   NetworkAnalyst                                                                                                                                               |
| <b>Programming Tools</b>           | R   Python                                                                                                                                                                   |
| <b>Machine Learning Tools</b>      | TensorFlow   Keras   NumPy   Pandas   Matplotlib                                                                                                                             |
| <b>Certificates</b>                | Machine Learning A-Z   Python for Data Science, AI & Development   Bulk RNA-Seq Data Analysis   Advanced Bioinformatics   Single-Cell RNA-Seq Data Analysis using R & Python |

## Research Experiences

**Project Intern at Nanyang Technological University, Singapore**

Jan 2026 - June 2026

- Developing a transcription factor activity inference pipeline on large-scale postmortem brain RNA-seq data to investigate gene regulatory dynamics across neurodegenerative disease progression.
- Supervised by Dr. Guillaume Thibault | Guillaume Thibault Lab

**Research Intern at Boston University, USA**

Sept 2025 - April 2026

- Developed an automated bioinformatics pipeline for transcription factor motif analysis across 200+ HPV genomes, identifying differentially enriched TFs between high-risk and low-risk types.
- Integrated motif-based approaches with deep learning sequence models to assess regulatory activity and characterise binding site conservation across 1,000+ clinical isolates.

**Research Intern at University of Liverpool, UK**

Sept 2025 - August 2026

- Analyzed single-cell RNA-seq data from CLL and Richter's Transformation patients, developing machine learning models (XGBoost, SVM, Random Forest) to differentiate disease states at the transcriptomic level.
- Built genome-scale metabolic models (GEMs) for CLL and Richter's Transformation cell types using the f1INIT algorithm, and identified differential metabolic reactions and enriched pathways between disease states and cell types.

**Research Intern at IIT Bombay, India**

May 2025 - Oct 2025

- Conducted differential expression analysis and WGCNA to identify key gene modules associated with autoimmune disease progression, followed by Gene Set Enrichment Analysis (GSEA) and Protein-Protein Interaction (PPI) network construction to uncover dysregulated pathways and functional gene relationships.
- Implemented predictive models to classify disease vs healthy states based on gene expression, enhancing biological interpretability through integrated network and machine learning approaches.

**Research Intern at Ashoka University, India**

Jan 2025 - Mar 2025

- Analyzed five assembled E. coli genomes for carbapenem and beta-lactam resistance using BLAST and RGI-CARD to identify mutations, and correlated genotype-phenotype relationships through AST report analysis.
- Performed Multiple Sequence Alignment (MSA) of Penicillin-Binding Proteins (PBPs) from the five E. coli genomes against a reference wild-type sequence to detect conserved mutations, and conducted genomic analysis that led to the identification of a 12-base pair insertion in the PBP3 gene, contributing to antibiotic resistance mechanisms.

**Biotech Team Lead at NovaLinks, India**

Feb 2025 - April 2025

- Led a project integrating miRNA analysis and histopathological imaging with machine learning to enhance lung cancer subtype classification and improve diagnostic precision.
- Conducted research on miRNA biomarkers specific to lung cancer and provided biological insights to support the development of an accurate classification model for distinguishing adenocarcinoma from squamous cell carcinoma.

## Publications

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**Comparative Analysis of Transcription Factor Binding Sites in the Long Control Region Across Human Papillomavirus Types** | *Viruses*, 2026, 18(6):646 (MDPI)

**Integrative transcriptomic analysis identifies predictive gene signatures across autoimmune diseases: implications for personalized therapeutic targeting** | *Under review at Journal of Autoimmunity* (Elsevier)

**Comparative Analysis of HPV-Positive Cervical and Head and Neck Cancers Using WGCNA and Machine Learning Identifies Shared Biomarkers** | *Under review at Tumour Virus Research* (Elsevier)

## Technical Projects

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### **Transcriptomic Comparison of HPV-Associated Cancers Using Co-Expression Networks**

- Conducted a comparative analysis of HPV-positive cervical and head and neck cancers using TCGA data. Performed DESeq2 and WGCNA to identify shared genes between both cancers.
- Applied machine learning models to filter features, identifying 16 HPV signature genes, and validated findings using single-cell RNA-seq analysis.
- Constructed PPI networks and performed functional enrichment analyses to identify shared molecular mechanisms underlying HPV-driven tumorigenesis.

### **Computational Identification of Antimalarial Leads through Fingerprint Similarity Analysis** [GitHub Link](#)

- Screened a library of antimalarial molecules against a query (chloroquine drug) by generating molecular fingerprints and computing Tanimoto similarity coefficients using KNIME and RDKit nodes.
- Extracted molecular data from the Protein Data Bank (PDB) and visualized pairwise similarity scores as a heatmap to identify high-similarity candidates for antimalarial drug development.

### **In Silico Analysis of 3DTC/CEP-6331 Complex using Molecular Docking** [GitHub Link](#)

- Implemented a molecular docking simulation using AutoDock Vina to model the interaction between the 3DTC kinase and the CEP-6331 inhibitor, achieving a predicted binding affinity of -12.23 kcal/mol.
- Analyzed the top binding pose to identify the specific amino acids responsible for the stable interaction, including key hydrophobic (ILE180, LEU274) and polar (LYS228, ASP284) residues.

### **Single-Cell Transcriptomic Profiling of the B16 Melanoma's Microenvironment** [GitHub Link](#)

- Processed raw single-cell transcriptomic data from a B16 melanoma model using R (Seurat), applying quality control, normalization, and clustering to map the tumor's cellular heterogeneity.
- Characterized the resulting cell clusters by annotating fourteen distinct cell types within the tumor microenvironment and performed differential expression analysis to identify their unique gene signatures.

## Honors and Leadership

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**First runner-up in the Biomimicry Innovation Challenge and Showcase**, Vellore Institute of Technology, 2024

**Top 10 Winner of Bio-Inspired Design Fest (BIDFEST) Ideathon**, Vellore Institute of Technology, 2024

### **Vice Chairperson at Alpha Bio Cell, VIT University, India** January 2024 - December 2024

- Led several biotechnology and bioinformatics-related events, including hackathons, workshops, and quizzes, enhancing the club's engagement.
- Mentored junior core members and applied leadership, technical, and project management skills to ensure successful event planning and execution.

## Extra-Curricular

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### **Volunteer | National Service Scheme, NGO**

*Conducted sessions in government schools on career opportunities and the importance of joining the armed forces.*

### **Volunteer | Becoming I Foundation, NGO**

*Taught science, math, and english in government schools, supporting underprivileged communities.*

### **Event Coordinator | BioSummit, School of Biosciences and Technology**

*Organized a flagship event connecting academia and food industries, managing food distribution and ensuring smooth execution.*